SPI Master Verification Test Plan

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
SPI_1_RESET	When PRESETn is asserted, all registers, FIFOs, status bits, SCLK, MOSI, and IRQ outputs should reset to default values.	Directed reset at simulation start	Cover registers reset values	scoreboard register reset values check
SPI_2_APB_RW	APB writes and reads should correctly access all SPI registers.	Directed APB transactions with specified addresses and random data.	Cover all register addresses and read/write accesses.	APB protocol assertions scoreboard register mirror checking.
SPI_3_APB_INVALID	Invalid or reserved APB accesses should not corrupt DUT registers.	directed invalid APB addresses.	Cover invalid/reserved register addresses and read/write accesses.	Checker verifies legal registers remain unchanged.
SPI_4_MODE_SWEEP	SPI master should support all 4 SPI modes (CPOL/CPHA).	Nested loops over modes 0,1,2,3 with randomized transfers and randomized mode selection.	Cross coverage: mode × width × bit-order.	Scoreboard checks sampled/transmitted data correctness. assertion checks SCLK idle level matches CPOL whenever BUSY=0 assertion checks Config stability during transfer .

SPI_5_WIDTH_SWEEP	SPI master should support 8-bit, 16-bit, and 32-bit transfers.	Randomized and directed width selection.	Cover all transfer widths.	Scoreboard checks RX data against expected values. assertion checks Config stability during transfer
SPI_6_BIT_ORDER	SPI should support both MSB-first and LSB-first operation.	Randomized and directed cfg_lsb_first configuration.	Cross coverage: width × lsb_first × mode.	Scoreboard checks RX data against expected values. assertion checks Config stability during transfer
SPI_7_CLK_DIV	SCLK period should follow CLK_DIV configuration.	Directed corner cases (0,1,31, FFFF) and randomized divider values.	Cover divider ranges: minimum, medium, maximum.	assertion checks Config stability during transfer assertion check for SCLK according to DIV
SPI_8_DELAY	Inter-transfer delay should follow DELAY register value.	Randomized DELAY values with back-to-back transfers.	Cover delay bins: zero, small, large.	Assertion check new transmitted data after delay passed checkers to check that Writes to DELAY during an active transfer take effect starting from the next transfer
SPI_9_SINGLE_TRANSFER	Single SPI transfer should complete correctly.	Directed single-transfer stimulus.	Cover transfer completion.	Scoreboard compares TX and RX values.
SPI_10_MULTI_TRANSFER	Consecutive SPI transfers should work correctly.	FIFO burst writes	Cover FIFO occupancy and back-to-back transfers.	Scoreboard verifies FIFO ordering
SPI_11_TX_FIFO	TX FIFO push/pop/full/empty behavior should be correct.	FIFO stress directed test.	Cover TX FIFO full/empty states	Assertions detect TX_FIFO overflow. checkers check TX_FIFO EMPTY/FULL flags
SPI_12_RX_FIFO	RX FIFO should correctly store received data.	FIFO stress directed test.	Cover RX FIFO full/empty states	Assertions detect RX_FIFO overflow. checkers check RX_FIFO EMPTY/FULL flags
SPI_13_LOOPBACK	Loopback mode should return	Enable cfg_loopback with directed data.	Cover loopback enabled/disabled.	Scoreboard compares TX and RX data.

	transmitted MOSI data on RX.			
SPI_14_SS_CONTROL	SS_n signals should assert/deassert correctly.	Directed SS_CTRL writes.		Immediate Assertions verify SS_n polarity Assertions verify SS_n held asserted for the entire WIDTH-bit transfer.
///SPI_15_BUSY_STATUS	busy signal should only assert during active transfers.	Randomized transfers with different delays/dividers.		Assertion checks busy behavior.
SPI_16_TRANSFER_DONE	transfer_done pulse should occur once per transfer.	Directed/randomized transfers.		Checkers verify interrupt state register updates
SPI_17_SCLK_IDLE	SCLK idle level should match CPOL whenever BUSY=0.	Randomized mode switching.		Assertion checks SCLK idle level matches CPOL.
SPI_18_MOSI_STABLE	MOSI should remain stable around non-launch edges.	Randomized SPI traffic.	Cover all SPI modes and edge combinations.	Assertion checks MOSI stable for at least 1 PCLK around each sample edge (WIRE-STABILITY). .
SPI_19_ENABLE_DISABLE	cfg_en=0 should disable transfers and force idle state.	directed enable toggling.		Assertions verify disabled-state behavior.
SPI_20_IRQ_GEN	IRQ should assert correctly for enabled interrupt sources	Directed interrupt triggering	Cover all interrupt sources	Immediate Assertion checks `IRQ == |(int_stat & int_en) Checkers verify TX_EMPTY/RX_FULL/TX_OVF/RX_OVF/DONE
SPI_21_IRQ_MASK	Masked interrupts should not assert IRQ	Directed/ Randomized Mask Sweep	Cover interrupt enable/mask combinations	
SPI_22_IRQ_W1C	Interrupt status bits should clear using W1C behavior	Directed W1C writes to INT_STAT	Cover interrupt clear operations	Checker verifies interrupt clearing